

Exp no:6 Determination and plotting of Capacitance of parallel plate capacitor in three different dielectric medium with specific area, dielectric thickness and various dielectric constant.

Date:

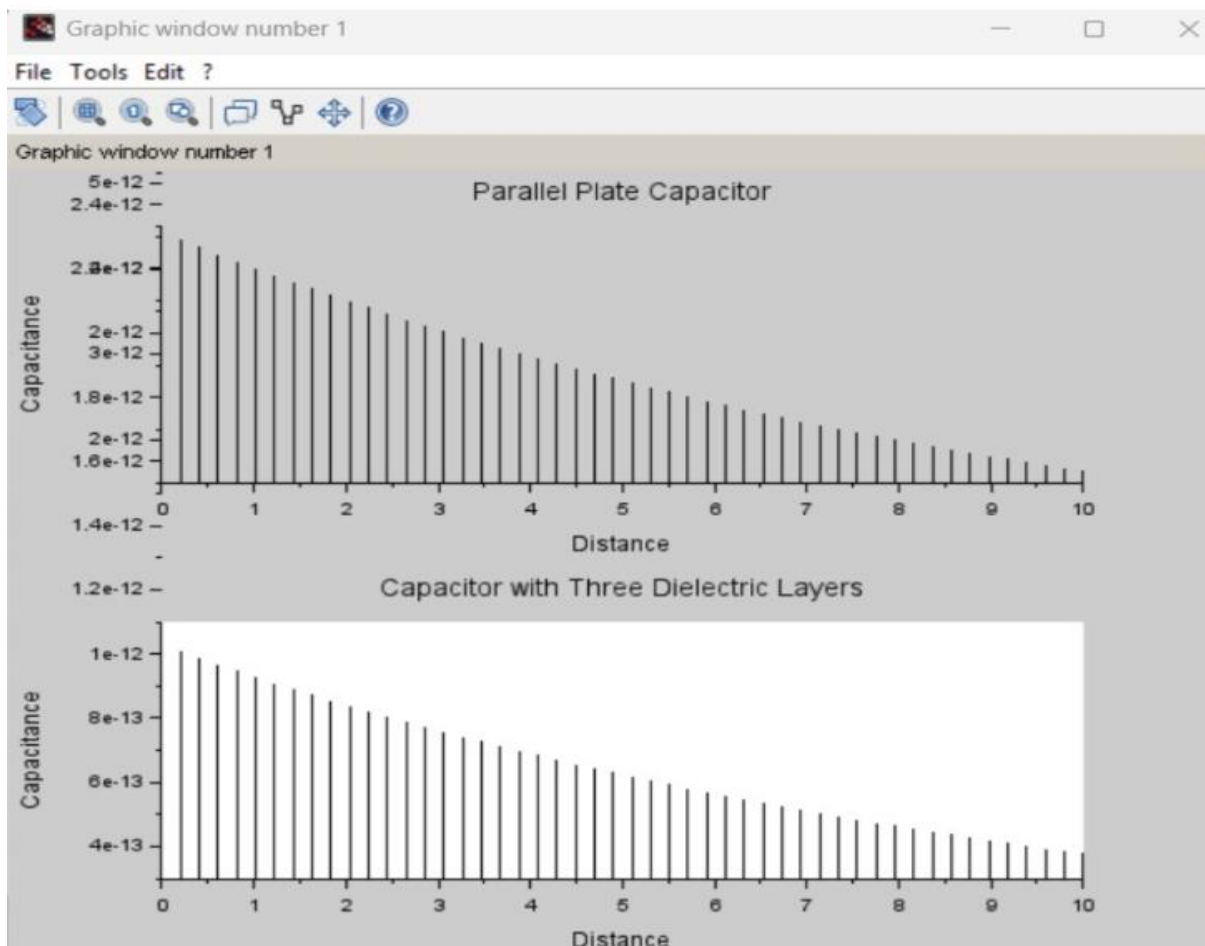
Aim:

To write program for capacitance of parallel plate capacitor in three different dielectric media using SCILAB.

Software required:

- SCILAB version 6.0.1

Partical caluction:



```
6-emf (C:\Users\vanac\6-emf) - Scilab
File Edit Format Options Window Execute ?
[Icons]
6-emf (C:\Users\vanac\6-emf) - Scilab
6-emf exp-7 exp-8
1 clc;
2 clear;
3 e = 8.854 * 10^-12;
4 a1 = input("Enter the plate area a1 (m^2): ");
5 d = input("Enter the distance d (m): ");
6 c = e * a1 / d;
7 disp(c, "Capacitance without dielectric (F) = ");
8 x = linspace(0, 10, 50);
9 y = c * exp(-0.1 * x);
10 figure();
11 subplot(2,1,1);
12 plot2d3(x, y);
13 xlabel("Distance");
14 ylabel("Capacitance");
15 title("Parallel Plate Capacitor");
16 a2 = input("Enter the plate area a2 (m^2): ");
17 d1 = input("Enter thickness d1 (m): ");
18 d2 = input("Enter thickness d2 (m): ");
19 d3 = input("Enter thickness d3 (m): ");
20 er1 = input("Enter relative permittivity er1: ");
21 er2 = input("Enter relative permittivity er2: ");
22 er3 = input("Enter relative permittivity er3: ");
23 c1 = e * a2 / ((d1/er1) + (d2/er2) + (d3/er3));
24 disp(c1, "Capacitance with dielectric (F) = ");
25 y1 = c1 * exp(-0.1 * x);
26 subplot(2,1,2);
```

Scilab 6.1.1 Console

File Edit Control Applications ?

Scilab 6.1.1 Console

```
Enter the plate area a1 (m^2): 0.0005
Enter the distance d (m): 1*10^-3

4.427D-12

"Capacitance without dielectric (F) = "
Enter the plate area a2 (m^2): 0.0004
Enter thickness d1 (m): 3*10^-3
Enter thickness d2 (m): 5*10^-3
Enter thickness d3 (m): 6*10^-3
Enter relative permittivity er1: 3
Enter relative permittivity er2: 4
Enter relative permittivity er3: 5

1.027D-12

"Capacitance with dielectric (F) = "
--> |
```

a) Area^(A₁) = 5 cm² $\epsilon_{r1} = 3$
 distance = 1 mm $\epsilon_{r2} = 4$
 Area^(A₂) = 4 cm² $\epsilon_{r3} = 5$
 $d_1 = 3$ mm
 $d_2 = 5$ mm
 $d_3 = 6$ mm

$$C = \frac{\epsilon_0 A}{d}$$

$$= \frac{(8.85 \times 10^{-12}) (0.005)}{1 \times 10^{-3}} = 4.427$$

$$C = \frac{\epsilon_0 A}{\left(\frac{d_1}{\epsilon_{r1}} \right) + \left(\frac{d_2}{\epsilon_{r2}} \right) + \left(\frac{d_3}{\epsilon_{r3}} \right)}$$

$$= \frac{(8.85 \times 10^{-12}) (0.005)}{\left(\frac{3 \times 10^{-3}}{3} \right) + \left(\frac{5 \times 10^{-3}}{4} \right) + \left(\frac{6 \times 10^{-3}}{5} \right)}$$

$$= 1.027 \times 10^{-12}$$

Results: Thus the program was verified for capacitance of parallel plate capacitor in three different dielectric media using SCILAB was successfully.